

Modal analysis and seismic response of steel frames with connection dampers

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Received 14 September 1999; received in revised form 11 April 2000; accepted 24 April 2000

Abstract

To reduce dynamic response of a steel frame of bolted connections, energy dissipation materials may be placed at a connection between the end plate and column flange or between the angle and member flange. By idealizing bolted connections and energy dissipation materials as rotational spring and damper respectively, this paper derives the mass matrix, stiffness matrix and damping matrix for the frame using a combination of the finite element method and the direct stiffness method. The complex modal analysis is then carried out to determine dynamic characteristics of the frame and to investigate the effects of connection stiffness and rotational damper on natural frequency and modal-damping ratio. By further introducing a generalized pseudo-excitation method, the vibration analysis of the frame subject to earthquake excitation is performed in the frequency domain to see how the connection stiffness and rotational damper affect the seismic performance of the frame. The parametric studies on the example frame with and without the connection dampers show that there is an optimal damper damping coefficient by which the modal-damping ratio of the frame can be considerably increased and the seismic responses, including both lateral displacements and internal forces, can be significantly reduced. © 2000 Elsevier Science Ltd. All rights reserved.

Keywords: Steel frame; Flexible connection; Rotational damper; Parametric study; Modal-damping ratio; Seismic response

1. Introduction

Most real beam-to-column connections used in steel frames are semi-rigid in that rotational discontinuity exists between connected beam and column. The analysis of steel frames adopting the assumption of rigid or pinned connections therefore may not be able to provide a good prediction of real behavior of a steel frame. Extensive research has been thus carried out in the past two decades to pursue the refined analysis and design of steel frames with semi-rigid connections [1–3]. Such efforts have led to provisions in various design codes of practice for semi-rigid connections [4,5].

The primary flexibility of a steel beam-to-column connection is its rotational deformation θ caused by the in-plane bending moment M . Based on extensive test results of various types of connections, a number of

mathematical models have been developed to describe the M – θ relationship of semi-rigid connections. These include linear model, bilinear model, piecewise linear model, cubic B-spline model, exponential model, polynomial model, and others. Correspondingly, the static analysis of semi-rigid jointed frames ranges from linear analysis, bifurcation analysis, semi-bifurcation analysis, and second order nonlinear analysis to ultimate analysis [1–3,6,7].

Compared with the static analysis, the research on dynamic analysis of semi-rigid jointed frames is relatively limited. Kawashima and Fujimoto [8] and Chan [9] investigated dynamic characteristics of steel frames with linear semi-rigid connections. Shi and Atluri [10], Al-Bermani et al. [11], Lui and Lopes [12], and Chui and Chan [13] examined the dynamic behavior of frames with nonlinear semi-rigid connections under cyclic and seismic loading in the time domain. They found that the rigidity of semi-rigid connections would influence the natural frequencies and dynamic response of the frames. The nonlinear semi-rigid connections would also absorb vibration energy.

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To reduce the seismic response of a steel frame with bolted connections, Hsu and Fafitis [14] proposed the elastomeric bolted connections by using two elastomeric pads and a shear pin as shown in Fig. 1(a). Some others just placed viscoelastic materials between the angles and beam flanges to remain the original bolted connection configuration unchanged [see Fig. 1(b)]. Such a bolted connection is usually modeled as a rotational spring and a rotational dashpot in parallel [14]. The previous study showed that the bolted connections with energy dissipating devices provided significant improvement by reducing the lateral displacement of the frame.

Further to Hsu and Fafitis' study, this paper derives the mass matrix, stiffness matrix and damping matrix for the beam element with rotational spring and damper at its ends using a combination of the finite element method and the direct stiffness method. The complex modal analysis is then carried out to determine the natural frequencies and modal damping ratios of the frame with the connection dampers. The seismic response of the frame is computed using a generalized pseudo-excitation method in the frequency domain. Parametric studies are performed to see the effects of connection stiffness and damper damping coefficient on the natural frequencies, modal damping ratios, and seismic response of the frame.

2. Formulation of element matrices

Fig. 2 shows a mechanical model for the beam with connection elements at its ends. The rotational spring provides the stiffness of connection while the rotational dashpot represents the energy dissipation capacity of connection dampers. By assuming that the beam is a Bernoulli beam and neglecting the effects of warping,

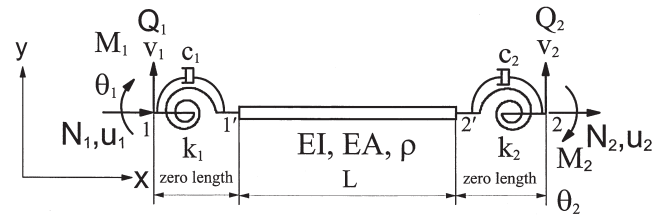


Fig. 2. Beam element with rotational spring and damper at its ends.

local buckling and cross sectional distortion, Chan [9] derived the stiffness and mass matrices for the beam with rotational springs only using the conventional finite element method. Kawashima and Fujimoto [8], on the other hand, derived the stiffness, mass, and damping matrices for the beam with both rotational springs and dampers using the direct stiffness method. The derivation of the element matrices using the direct stiffness method is rather complicated and difficult to be understood by engineers. However, it is also difficult to derive the damping matrix for the beam of rotational dampers by using the finite element method. This paper thus combines the finite element method with the direct stiffness approach to derive the element matrices.

For the beam element with rotational springs and dashpots at its ends as shown in Fig. 2, M_j and θ_j ($j=1,2$) are the moment and rotational displacement at the global node j while $M_{j'}$ and $\theta_{j'}$ are the moment and rotational displacement at the internal node j' . Since the connection element is of zero length, the axial force and shear force and the corresponding displacements at the global node j are the same as those at the internal node j' . Thus, only moments M and rotational displacements θ are needed to be distinguished between global nodes j and internal nodes j' ($j=1,2$). Based on the results from the finite element method, the dynamic stiffness matrix relating

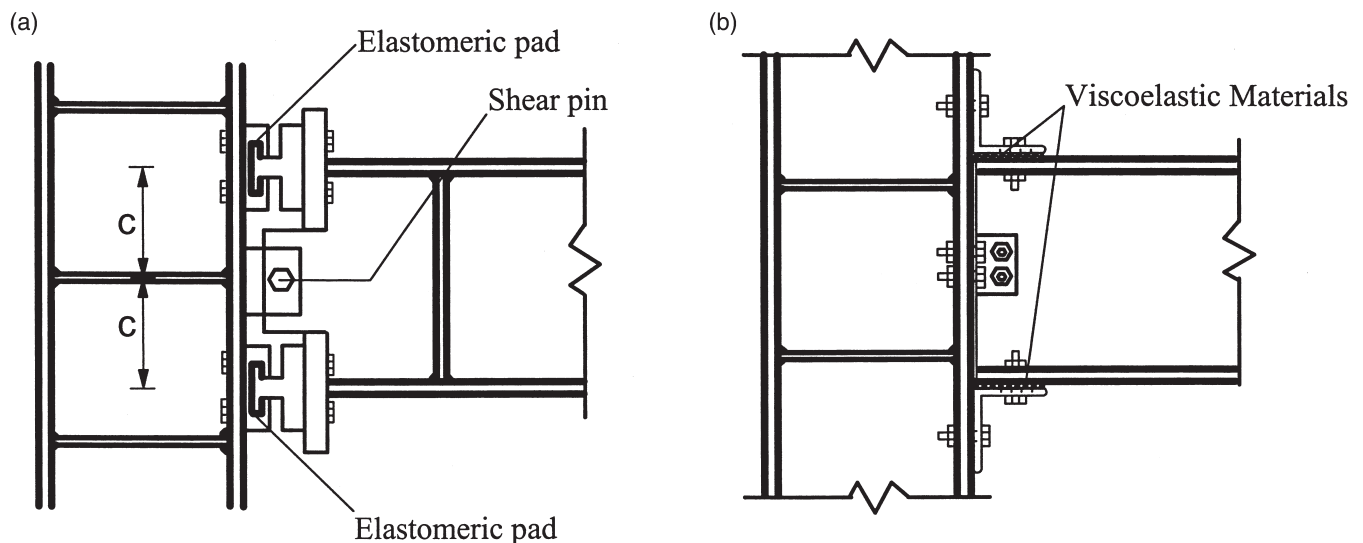


Fig. 1. Beam-column connection with energy dissipation devices. (a) Elastomeric pad [14]. (b) Viscoelastic materials.

harmonic displacement vector $\mathbf{d}$ and harmonic force vector $\mathbf{f}$ at the internal nodes of the beam element can be written as

$$\mathbf{f} = \mathbf{D} \mathbf{d} \quad (1)$$

$$\mathbf{d} = [u_1, v_1, \theta'_1, u_2, v_2, \theta'_2]^T \quad (2)$$

$$\mathbf{f} = [N_1, Q_1, M'_1, N_2, Q_2, M'_2]^T \quad (3)$$

in which $\mathbf{D} = [D_{ij}] = \mathbf{K}_b - \omega^2 \mathbf{M}_b$, a 6×6 symmetric dynamic stiffness matrix; ω is the frequency in rad/s. $\mathbf{K}_b$ and $\mathbf{M}_b$ are the conventional stiffness and consistent mass matrices, respectively, of the beam element in the finite element method (see Appendix A).

The dynamic stiffness matrix relating harmonic rotational displacements and harmonic moments of the connection shown in Fig. 2 can be written as

$$\begin{Bmatrix} M_j \\ M'_j \end{Bmatrix} = \begin{bmatrix} r_j & -r_j \\ -r_j & r_j \end{bmatrix} \begin{Bmatrix} \theta_j \\ \theta'_j \end{Bmatrix} \quad (j=1,2) \quad (4)$$

in which $r_j = k_j + i\omega c_j$; r_j is the dynamic stiffness of the connection; k_j and c_j are the rotational stiffness and rotational damper damping coefficient of the connection; and i is the imaginary unit.

The assemblage of the beam element and the two connections based on Eqs. (1) and (4) leads to the following relationship.

$$\begin{Bmatrix} M_1 \\ M'_1 \\ M_2 \\ M'_2 \\ Q_1 \\ Q_2 \\ N_1 \\ N_2 \end{Bmatrix} = \begin{bmatrix} r_1 & -r_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -r_1 & r_1 + D_{33} & D_{36} & 0 & D_{32} & D_{35} & D_{31} & D_{34} \\ 0 & D_{63} & r_2 + D_{66} & -r_2 & D_{62} & D_{65} & D_{61} & D_{64} \\ 0 & 0 & -r_2 & r_2 & 0 & 0 & 0 & 0 \\ 0 & D_{23} & D_{26} & 0 & D_{22} & D_{25} & D_{21} & D_{24} \\ 0 & D_{53} & D_{56} & 0 & D_{52} & D_{55} & D_{51} & D_{54} \\ 0 & D_{13} & D_{16} & 0 & D_{12} & D_{15} & D_{11} & D_{14} \\ 0 & D_{43} & D_{46} & 0 & D_{42} & D_{45} & D_{41} & D_{44} \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta'_1 \\ \theta_2 \\ \theta'_2 \\ v_1 \\ v_2 \\ u_1 \\ u_2 \end{Bmatrix} \quad (5)$$

If the external loads are applied at the global nodes only, M'_1 and M'_2 in Eq. (5) should be zero, resulting in

$$\begin{Bmatrix} \theta'_1 \\ \theta'_2 \end{Bmatrix} = \frac{1}{\beta} \begin{bmatrix} r_2 + D_{66} & -D_{36} \\ -D_{63} & r_1 + D_{33} \end{bmatrix} \begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} - \begin{bmatrix} D_{32} & D_{35} & D_{31} & D_{34} \\ D_{62} & D_{65} & D_{61} & D_{64} \end{bmatrix} \begin{Bmatrix} v_1 \\ v_2 \\ u_1 \\ u_2 \end{Bmatrix} \quad (6)$$

in which $\beta = (r_1 + D_{33})(r_2 + D_{66}) - D_{36}D_{63}$.

It is seen that the rotational displacements at the internal nodes can be expressed in terms of displacements at the global nodes. As a result, Eq. (5) can be reduced to a 6×6 dynamic stiffness matrix in terms of the global nodes only. This can be achieved by substituting Eq. (6) into Eq. (5) and some manipulation afterwards.

$$\begin{Bmatrix} M_1 \\ M_2 \\ Q_1 \\ Q_2 \\ N_1 \\ N_2 \end{Bmatrix} = \begin{bmatrix} [f_1]_{2 \times 2} & [f_2]_{2 \times 4} \\ [f_3]_{4 \times 2} & [f_4]_{4 \times 4} \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \\ v_1 \\ v_2 \\ u_1 \\ u_2 \end{Bmatrix} = \mathbf{Z} \begin{Bmatrix} \theta_1 \\ \theta_2 \\ v_1 \\ v_2 \\ u_1 \\ u_2 \end{Bmatrix} \quad (7)$$

in which

$$[f_1]_{2 \times 2} = \begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix} \quad (8)$$

$$[f_2]_{2 \times 4} = \begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix} \begin{bmatrix} r_2 + D_{66} & -D_{36} \\ -D_{63} & r_1 + D_{33} \end{bmatrix} \begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix} - \frac{1}{\beta} \begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix} \begin{bmatrix} D_{32} & D_{35} & D_{31} & D_{34} \\ D_{62} & D_{65} & D_{61} & D_{64} \end{bmatrix} \quad (9)$$

$$[f_3]_{4 \times 2} = \begin{bmatrix} D_{22} & D_{25} & D_{21} & D_{24} \\ D_{52} & D_{55} & D_{51} & D_{54} \\ D_{12} & D_{15} & D_{11} & D_{14} \\ D_{42} & D_{45} & D_{41} & D_{44} \end{bmatrix} \quad (10)$$

$$[f_4]_{4 \times 4} = \begin{bmatrix} D_{23} & D_{26} \\ D_{53} & D_{56} \\ D_{13} & D_{16} \\ D_{43} & D_{46} \end{bmatrix} \begin{bmatrix} r_2 + D_{66} & -D_{36} \\ -D_{63} & r_1 + D_{33} \end{bmatrix} \begin{bmatrix} D_{32} & D_{35} & D_{31} & D_{34} \\ D_{62} & D_{65} & D_{61} & D_{64} \end{bmatrix} - \frac{1}{\beta} \begin{bmatrix} D_{23} & D_{26} \\ D_{53} & D_{56} \\ D_{13} & D_{16} \\ D_{43} & D_{46} \end{bmatrix}$$

By expanding the elements of $\mathbf{Z}$ in series with respect to the circular frequency ω and neglecting higher terms than the third order, the following expression is obtained

$$\mathbf{Z} = \mathbf{K}_e + i\omega\mathbf{C}_e - \omega^2\mathbf{M}_e \quad (11)$$

$\mathbf{K}_e$, $\mathbf{C}_e$, and $\mathbf{M}_e$ are the required stiffness matrix, damping matrix and mass matrix of the hybrid element shown in Fig. 2, respectively. $\mathbf{K}_e$, $\mathbf{C}_e$, and $\mathbf{M}_e$ can be obtained in an explicit form, as listed in Appendix A, and they are all the functions of the fixity factor ν defined as [6]

$$k = \frac{3EI}{L} \frac{\nu}{1-\nu} \quad (12)$$

It is seen that the fixity factor ν relates the rotational stiffness of the connection to the flexural rigidity of the beam. For a pinned connection, the fixity factor equals zero and for a rigid connection it equals one. For the connection type shown in Fig. 1(a), the fixity factor is above zero but below one, determined by the rotational stiffness of the connection k that depends on the elastic modulus, area, and thickness of the elastomeric material. The details can be found in Hsu and Fafitis [14]. Apart from the fixity factor, the damping matrix and mass matrix $\mathbf{C}_e$ and $\mathbf{M}_e$ are also the functions of the damper damping coefficient.

3. Solution for dynamic characteristics

In this study, the rotational springs at connections are assumed to be linear and elastic and all members of the frame are in the linear and elastic range under earthquake excitation. The energy dissipation capacity of the bolted connection itself is considered to be included in the rotational damper already. These assumptions indicate that the considered frame experiences only moderate or minor earthquake excitations, or because of the significant increase of energy dissipation capacity due to the installation of rotational dampers the frame is able to remain linear and elastic during an earthquake event. With these assumptions and by using the finite element method and the derived element matrices, the equation of motion of a flexibly jointed frame with rotational dampers under seismic excitation (see Fig. 3) can be expressed as

$$\mathbf{M}\ddot{\mathbf{X}}(t) + \mathbf{C}\dot{\mathbf{X}}(t) + \mathbf{K}\mathbf{X}(t) = -\mathbf{M}\mathbf{E}\ddot{x}_g(t) \quad (13)$$

where $\mathbf{M}$ and $\mathbf{K}$ are the mass matrix and stiffness matrix of the frame of flexible connections, respectively; $\mathbf{C}$ is the total damping matrix of the frame, including structural damping as well as the damping from the rotational dampers; $\mathbf{X}(t)$ is the vector of relative displacement response with respect to the ground; $\ddot{x}_g(t)$ is the ground acceleration and $\mathbf{E}$ is the index vector given by

$$\mathbf{E} = [1, 0, 0, 1, 0, 0, \dots, 1, 0, 0]^T \quad (14)$$

Clearly, the semi-rigid frame with connection dampers is a non-classically damped system. To facilitate the

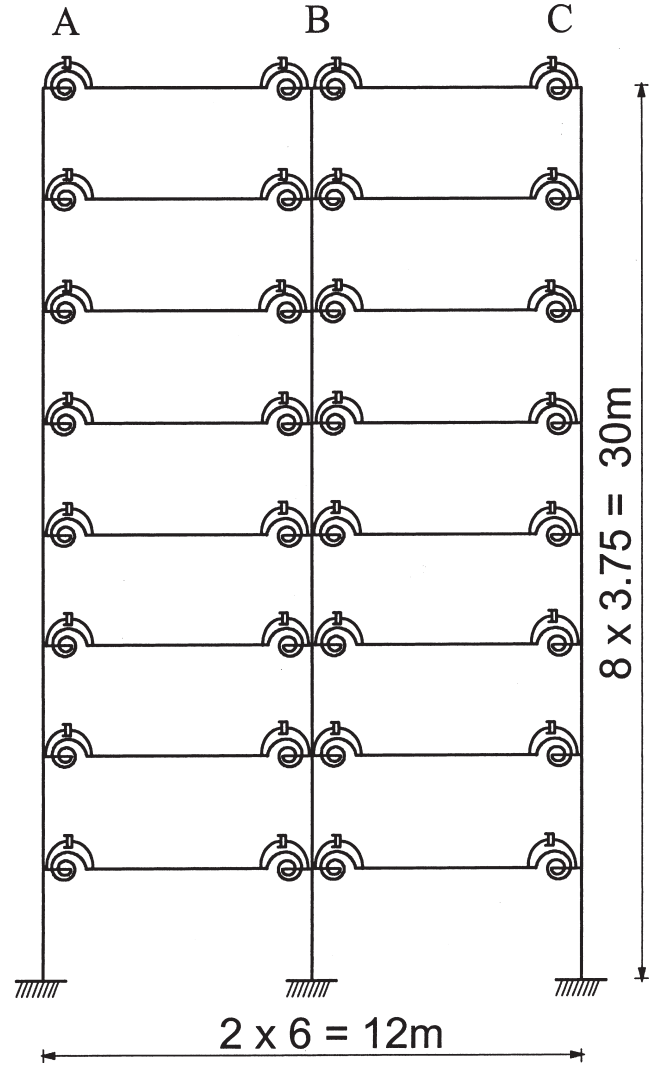


Fig. 3. Steel frame with rotational dampers at flexible joints.

seismic design of the frame, the complex mode superposition method is employed in this study to estimate the modal damping ratios and natural frequencies.

Eq. (13) can be replaced by the first-order 2N-dimensional equation of the form

$$\mathbf{A}\dot{\mathbf{Y}} + \mathbf{B}\mathbf{Y} = \mathbf{F}\ddot{x}_g(t) \quad (15)$$

where

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{M} \\ \mathbf{M} & \mathbf{C} \end{bmatrix}, \mathbf{B} = \begin{bmatrix} -\mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{K} \end{bmatrix}, \mathbf{F} = \begin{Bmatrix} \mathbf{0} \\ -\mathbf{M}\mathbf{E} \end{Bmatrix} \text{ and } \mathbf{Y} = \begin{Bmatrix} \dot{\mathbf{X}} \\ \mathbf{X} \end{Bmatrix} \quad (16)$$

If the solution of the homogeneous form of Eq. (13) is taken as

$$\mathbf{X}(t) = \phi e^{st} \quad (17)$$

in which s is the eigenvalue and ϕ is the associated eigenvector, the solution of the homogeneous form of Eq. (15) can be then taken as

$$\mathbf{Y} = \begin{Bmatrix} s\phi \\ \phi \end{Bmatrix} e^{st} = \hat{\phi} e^{st} \quad (18)$$

The associated eigenvalue problem becomes

$$(s\mathbf{A} + \mathbf{B})\hat{\phi} = \mathbf{0} \quad (19)$$

3.1. Underdamped system

The solution of Eq. (19) comprises a set of $2N$ eigenvalues and eigenvectors that are real or exist in complex conjugate pairs. If all the eigenvalues and eigenvectors are in complex conjugate pairs with negative real parts, the system is defined as the underdamped system. Thus, for an underdamped system, the eigenvectors and eigenvalues are

$$\hat{\phi}_j = \hat{\phi}_{j+N}^* \text{ and } s_j = s_{j+N}^* \quad (j=1,2,\dots,N) \quad (20)$$

Each eigenvalue can be written under the form

$$s_j = s_{j+N}^* = -\omega_j \xi_j + i\omega_{dj} \quad (j=1,2,\dots,N) \quad (21)$$

in which

$$\omega_j = |s_j|, \quad \xi_j = -\text{Re}(s_j)/|s_j| \text{ and } \omega_{dj} = \omega_j \sqrt{1 - \xi_j^2} \quad (22)$$

ω_j , ω_{dj} , and ξ_j are the modal frequency, the damped modal frequency, and the modal damping ratio, respectively, associated with mode j . The superscript $*$ means the conjugation. For the underdamped system, the value of ξ_j is always less than one.

3.2. System with mixed damping

For a semi-rigid frame with connection dampers, Eq. (19) may produce some real-valued negative pairs, each associated with a real-valued eigenvector. This is because when rotational damper damping coefficient becomes very large some higher modes of vibration of the frame will be damped out. In this case, the system may be called the mixed damped system because some of the eigenvalues and eigenvectors are still in complex conjugate pairs with negative real parts. It is convenient to express real pairs s_j in the following form analogous to Eq. (21).

$$s_j = -\omega_j \xi_j + \omega_{dj} \quad (j=1,2,\dots,N) \quad (23)$$

$$s_{j+N} = -\omega_j \xi_j - \omega_{dj} \quad (j=1,2,\dots,N)$$

in which ω_j , ω_{dj} and ξ_j are determined by

$$\omega_j = \sqrt{s_j s_{j+N}}, \quad \xi_j = -(s_j + s_{j+N})/(2\omega_j) \text{ and } \omega_{dj} \quad (25)$$

$$= \omega_j \sqrt{\xi_j^2 - 1} = (s_j - s_{j+N})/2$$

The value of ξ_j is now equal to or greater than one.

4. Solution for seismic response

The complex mode superposition method adopts the following coordinate transformation to decouple Eq. (15)

$$\mathbf{Y} = \Phi \mathbf{Z} \quad (26)$$

where $\mathbf{Z}$ is the $2N$ -dimensional generalized coordinate vector and Φ is the $2N \times 2N$ complex modal matrix.

$$\Phi = [\hat{\phi}_1, \hat{\phi}_2, \dots, \hat{\phi}_{2N}] \quad (27)$$

By using the coordinate transformation and the orthogonality of modes, Eq. (15) can be reduced to $2N$ decoupled modal equations with the j th modal equation being

$$A_j \dot{Z}_j + B_j Z_j = \hat{\phi}_j^T \mathbf{F} \ddot{X}_g(t) \quad (28)$$

or alternatively

$$\dot{Z}_j - s_j Z_j = r_j \ddot{X}_g(t) \quad (29)$$

in which

$$B_j = \hat{\phi}_j^T \mathbf{B} \hat{\phi}_j = -s_j A_j \quad (30)$$

$$r_j = \hat{\phi}_j^T \mathbf{F} / A_j = -\hat{\phi}_j^T \mathbf{M} \mathbf{E} / A_j \quad (31)$$

Assume that the ground acceleration $\ddot{X}_g(t)$ is a stationary random process and its power spectral density function is given as $S_g(\omega)$. A generalized pseudo-excitation method in conjunction with the complex mode superposition method are now developed to determine the seismic response of a semi-rigid frame with connection dampers.

The pseudo-excitation is constituted for a given frequency ω as

$$\ddot{X}_g(t) = \sqrt{S_g(\omega)} e^{i\omega t} \quad (32)$$

The solution of the first-order Eq. (29) to the pseudo-excitation is

$$Z_j(\omega, t) = \frac{r_j}{i\omega - s_j} \sqrt{S_g(\omega)} e^{i\omega t} \quad (j=1,2,\dots,2N) \quad (33)$$

Substituting Eq. (33) into Eq. (26) and comparing with the last part of Eq. (16), one obtains

$$\mathbf{X}(\omega, t) = \sum_{j=1}^{2N} \phi_j Z_j(\omega, t) = \sum_{j=1}^{2N} \phi_j \frac{r_j}{i\omega - s_j} \sqrt{S_g(\omega)} e^{i\omega t} \quad (34)$$

Since the eigenvector is in pairs in either underdamped system or the system with mixed damping, the $\mathbf{X}(\omega, t)$ can be reduced to

$$\mathbf{X}(\omega, t) = \sum_{j=1}^N H_j(\omega) (i\omega \alpha_j + \beta_j) \sqrt{S_g(\omega)} e^{i\omega t} = \mathbf{X}(\omega) e^{i\omega t} \quad (35)$$

in which $\mathbf{X}(\omega)$ is called the pseudo displacement

$$\mathbf{X}(\omega) = \sum_{j=1}^N H_j(\omega)(i\omega\alpha_j + \beta_j)\sqrt{S_g(\omega)} \quad (36)$$

$H_j(\omega)$ is the frequency response function for the j th mode.

$$H_j(\omega) = \frac{1}{\omega_j^2 - \omega^2 + i2\xi_j\omega\omega_j} \quad (37)$$

When the j th mode is a underdamped mode,

$$\alpha_j = 2\text{Re}(\phi_j r_j) \text{ and } \beta_j = -2\text{Re}(\phi_j r_j s_j^*) \quad (38)$$

When the j th mode is an overdamped mode or a critically damped case

$$\alpha_j = (\phi_j r_j + \phi_{j+N} r_{j+N}) \text{ and } \beta_j = -(\phi_j r_j s_{j+N} + \phi_{j+N} r_{j+N} s_j) \quad (39)$$

Similarly, the pseudo velocity response can be obtained by

$$\dot{\mathbf{X}}(\omega) = \sum_{j=1}^N H_j(\omega)(i\omega\mu_j + \nu_j)\sqrt{S_g(\omega)} \quad (40)$$

in which when the j th mode is a underdamped mode

$$\mu_j = 2\text{Re}(s_j \phi_j r_j) \text{ and } \nu_j = -2\omega_j^2 \text{Re}(\phi_j r_j) \quad (41)$$

When the j th mode is an overdamped mode or a critically damped case

$$\mu_j = (s_j \phi_j r_j + s_{j+N} \phi_{j+N} r_{j+N}) \text{ and } \nu_j = -\omega_j^2 (\phi_j r_j + \phi_{j+N} r_{j+N}) \quad (42)$$

Once the pseudo displacement is determined, the pseudo internal force, $\mathbf{Q}(\omega)$, can be easily determined following a static analysis.

The response spectral matrix can be then obtained by

$$\mathbf{S}_{xx}(\omega) = \mathbf{X}^*(\omega)\mathbf{X}^T(\omega), \quad \mathbf{S}_{x\dot{x}}(\omega) = \mathbf{X}^*(\omega)\dot{\mathbf{X}}^T(\omega) \quad (43)$$

$$\mathbf{S}_{\dot{x}\dot{x}}(\omega) = \dot{\mathbf{X}}^*(\omega)\dot{\mathbf{X}}^T(\omega), \quad \mathbf{S}_{\dot{x}x}(\omega) = \dot{\mathbf{X}}^*(\omega)\mathbf{X}^T(\omega) \quad (44)$$

$$\mathbf{S}_{QQ}(\omega) = \mathbf{Q}^*(\omega)\mathbf{Q}^T(\omega) \quad (45)$$

The standard deviation displacement response of the j th floor σ_{x_j} is finally evaluated from $S_{x_j x_j}(\omega)$ through integration.

$$\sigma_{x_j}^2 = \int_{-\infty}^{+\infty} S_{x_j x_j}(\omega) d\omega \quad (46)$$

The standard deviation acceleration response of the j th floor $\sigma_{\ddot{x}_j}$ is given by

$$\sigma_{\ddot{x}_1}^2 = \int_{-\infty}^{+\infty} \omega^2 S_{\ddot{x}_j \ddot{x}_j}(\omega) d\omega \quad (47)$$

5. Numerical example

An eight story two bay semi-rigid frame with rotational dampers is selected as an example building (see Fig. 3) for parametric study and application. The material properties of the frame members are given in Table 1. The structural damping ratio is set to 1% for the first two vibration modes of the frame without rotational dampers, following the assumption of Rayleigh damping. The stiffness of rotational spring and the damping coefficient of rotational damper are the same for all connections. The Kanai-Tajimi filtered white noise spectrum is used as the ground acceleration spectrum.

$$S_g(\omega) = \frac{1 + 4\xi_g^2 \left(\frac{\omega}{\omega_g}\right)^2}{\left[1 - \left(\frac{\omega}{\omega_g}\right)^2\right]^2 + 4\xi_g^2 \left(\frac{\omega}{\omega_g}\right)^2} S_0 \quad (48)$$

in which ω_g , ξ_g , S_0 may be regarded as the characteristics and the intensity of an earthquake in a particular geological location. The parameters in the ground acceleration spectrum are selected as $\omega_g = 15.0$ rad/s; $\xi_g = 0.65$ and $S_0 = 4.65 \times 10^{-4}$ m²/rad²•s³.

5.1. Modal frequencies and modal damping ratios

The modal frequencies of the frame without rotational dampers are computed first in this study. Fig. 4 displays the variations of the first two modal frequencies (n_1 and n_2) of the frame with the fixity factor. Clearly, the stiffness of rotational springs affects the modal frequencies of the frame significantly. The smaller rotational stiffness (i.e. the smaller fixity factor) leads to the smaller modal frequencies. When the fixity factor is unit, which represents the frame of rigid connections, the first and second modal frequencies, n_1 and n_2 , are 2.78 and 8.73 Hz respectively. As the fixity factor reduces to 0.2, the first and second modal frequencies become 1.26 and 4.48 Hz. It is interesting to see that the relationship between the modal frequency and the fixity factor is almost linear but this may not be true for other frames.

The modal damping ratios of the semi-rigid frame with rotational dampers for a set of fixity factors are then computed against damper damping coefficient. The computed first and second modal damping ratios of the frame are shown in Fig. 5(a,b). For the fixity factor equal to unit, no matter what value is the damper damping

Table 1
Material properties of members in example frame

Member	EI (N•m ²)	EA (N)	ρA (kg/m)
Beam	1.553×10^8	2.645×10^9	101.2
Column	2.048×10^8	7.503×10^9	287.1

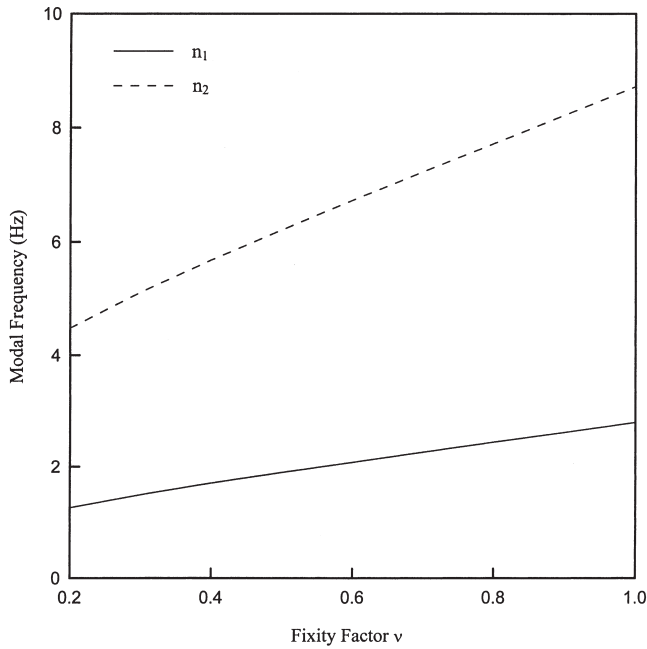


Fig. 4. Variations of modal frequencies with fixity factor (no damper case).

coefficient, the first and second modal damping ratios of the frame remain 0.01, i.e. the structural damping ratios set for the frame without rotational dampers. This result is expected because when the connections become rigid the rotational dampers lose their functions. For the frame of flexible connections, it is seen from Fig. 5(a,b) that if the damper damping coefficient is small, the modal damping ratios of the frame are almost unchanged. Only when the damper damping coefficient reaches above a certain value, the modal damping ratios are significantly increased first and then decreased as the damper damping coefficient becomes even larger. The dampers of very large damper damping coefficient will lead to very small rotational displacements and velocities of the connections and, as a result, much less vibration energy can be absorbed by the rotational dampers. It is clear from Fig. 5(a,b) that there is an optimal damper damping coefficient for a given fixity factor by which the modal damping ratio of the frame is the maximum. The optimal damper damping coefficient and corresponding maximum modal-damping ratio, however, vary with the fixity factor, i.e. the connection stiffness.

The parametric study for modal properties is finally moved to the optimal damper damping coefficient and the corresponding maximum modal-damping ratio of the frame. Fig. 6(a) shows the variations of the optimal damper damping coefficient and the corresponding maximum modal damping ratio with the stiffness of the rotational springs for the first mode of vibration of the frame. Fig. 6(b) displays the variations of the corresponding first damped modal frequency with the stiffness of rotational springs. As the flexible connections become

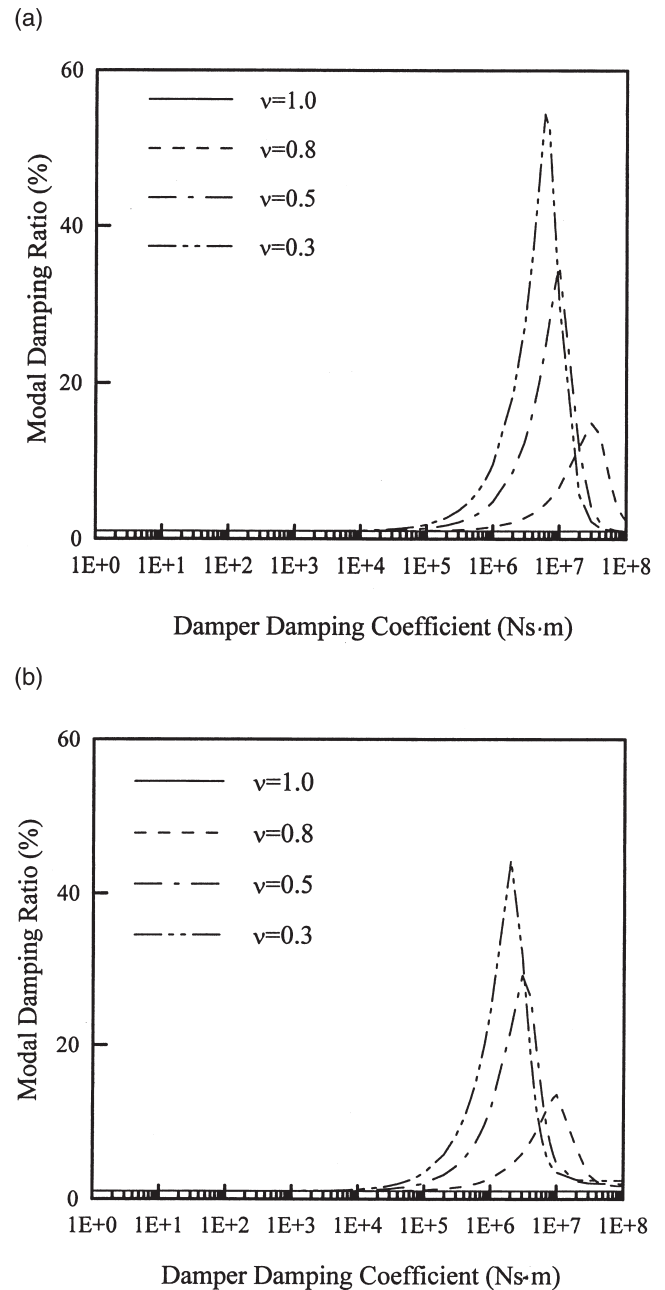


Fig. 5. Variations of modal damping ratio with damper damping coefficient. (a) First modal damping ratio. (b) Second modal damping ratio.

more and more stiff, the optimal damper damping coefficient required for the maximum modal damping ratio of the frame becomes larger and larger, but the achievable maximum modal damping ratio becomes smaller and smaller. Although the first damped modal frequency of the frame still increases with the increasing stiffness of the connections, the relationship between the two quantities is no longer approximately linear. Also because of the rotational dampers of optimal value, the first damped modal frequency becomes larger if compared with that of the frame without rotational dampers (see Fig. 4).

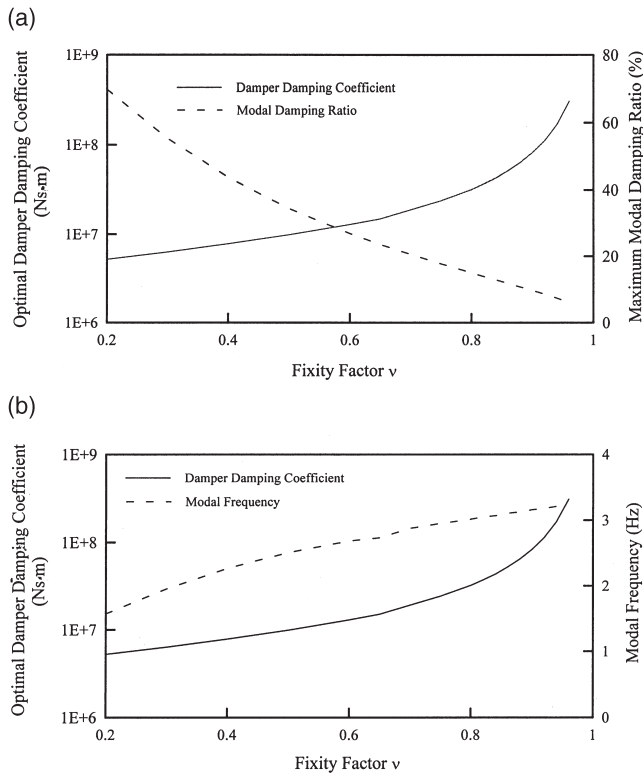


Fig. 6. Variations of optimal modal properties with fixity factor. (a) Optimal damper damping coefficient and maximum modal damping ratio. (b) Optimal damper damping coefficient and modal frequency.

From the above analysis, it is understood that both the connection stiffness and damper damping coefficient affect the modal frequencies and modal damping ratios of the frame. For a given connection stiffness and a given mode of vibration, there exists an optimum value of damper damping coefficient by which the modal damping ratio of the frame reaches its maximum. The high modal-damping ratio generated by the rotational dampers of optimal value is expected to significantly bring down the seismic response of the frame.

5.2. Seismic responses

Seismic response analysis is carried out to investigate the variations of seismic response of the frame with connection stiffness and rotational damper damping coefficient and to see if the optimal modal parameters identified from the modal analysis are the same as those from the seismic response analysis. Then, the effectiveness of optimal modal parameters on seismic response reduction is examined.

Fig. 7 depicts the variations of the standard deviation of top displacement response of the frame at Column C (see Fig. 3) with damper damping coefficient for several connection fixity factors. When the damper damping coefficient is relatively small, it does not affect the standard deviation of top displacement response but the less

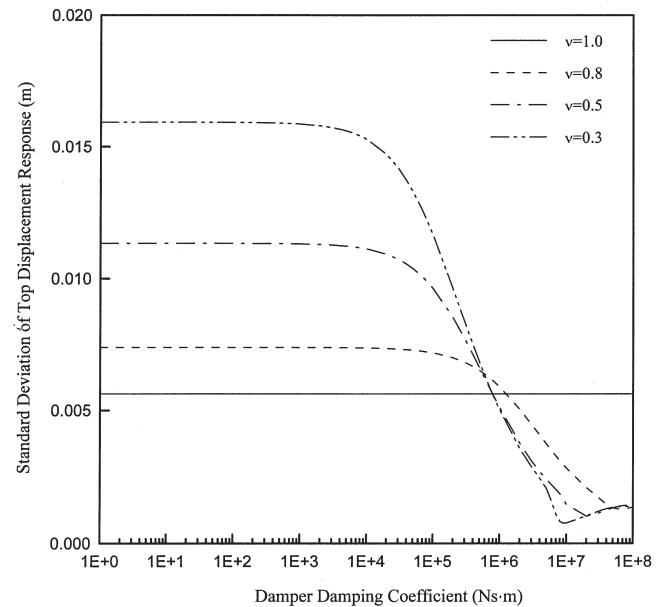


Fig. 7. Top displacement response of frame at column C vs damper damping coefficient.

the connection stiffness (the fixity factor) is, the larger the standard deviation of top displacement response. For a given fixity factor, when the damper damping coefficient increases to a certain level it starts to significantly reduce the displacement response of the frame with flexible connections. This level depends on the fixity factor: the smaller the fixity factor, the smaller the level of damper damping coefficient. This is consistent with the results from the modal analysis as shown in Fig. 5. In particular, when the damper damping coefficient is further increased to be above 1×10^6 Nsm the top displacement response of the frame with flexible connections becomes less than that of the frame with rigid connection. This is because of the high modal damping ratios attributed from rotational dampers.

Fig. 8 shows the variations of the standard deviation of base shear force of the frame at Column C with damper damping coefficient for several connection fixity factors. It is seen that the smaller connection fixity factor can result in the smaller base shear force response of the frame even without rotational dampers. With rotational dampers and when damper damping coefficient is over a certain level the base shear force response can be further reduced. When the damper damping coefficient reaches its optimal value identified from the modal analysis, the base shear force response of the frame becomes the smallest. For instance, the standard deviation of base shear force of the frame at Column C with perfectly rigid connections is 10,784 N. With the flexible connections of fixity factor $\nu=0.3$ and damper damping coefficient $c=3 \times 10^6$ Nsm, it is reduced to 2234 N, leading to a 79% reduction of the base shear response. However, for the frame of small fixity factor, the high modes of vibration

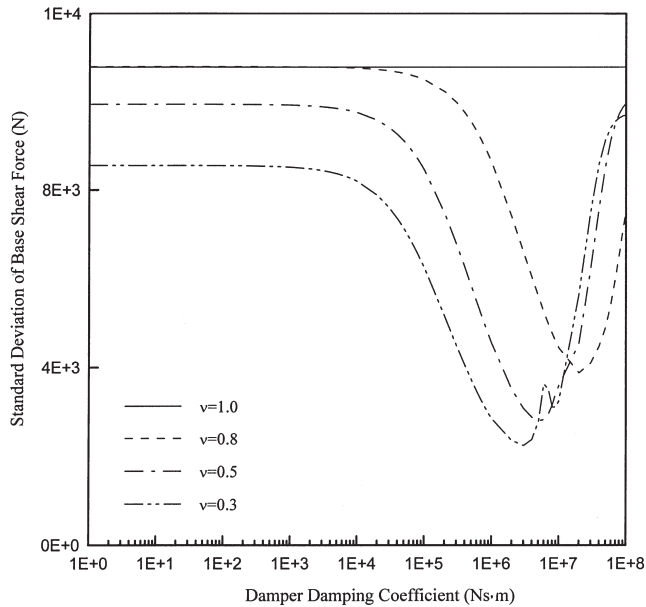


Fig. 8. Base shear force response of frame at column C vs damper damping coefficient.

also make some contributions to the response near the optimal damper damping coefficient, as evidenced by the small peak in Fig. 8.

Fig. 9 shows the variations of the standard deviation of base bending moment of the frame at Column C with damper damping coefficient for several connection fixity factors. It is seen that the magnitude of bending moment response depends on the value of fixity factor. Without rotational dampers or with rotational dampers but of smaller damper damping coefficient, the smaller fixity

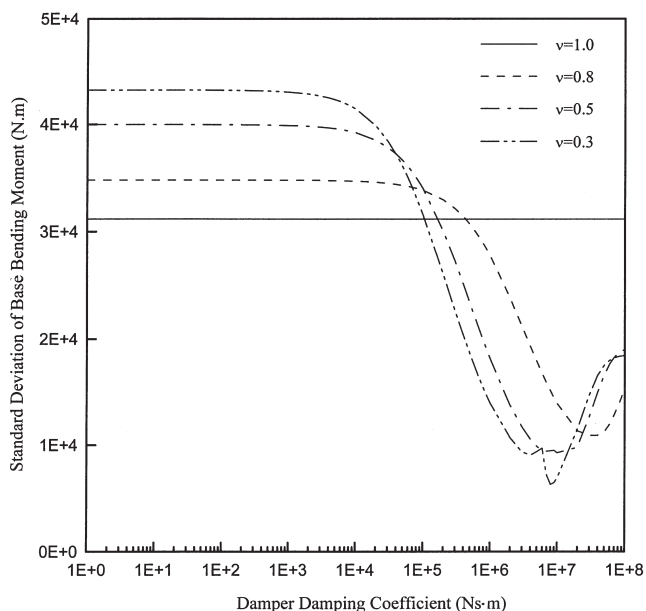


Fig. 9. Base bending moment response of frame at column C vs damper damping coefficient.

factor leads to the larger base bending moment. This may be because the base connections are rigid and the decrease in stiffness of frame due to connection flexibility causes the beam-to-column connections to shed their moments to the base connections. It is also seen from Fig. 9 that when the damper damping coefficient is increased to a certain level, the further increase of damper damping coefficient will reduce the bending moment response significantly. The minimum bending moment response can be achieved at the optimal damper damping coefficient identified from the modal analysis. The standard deviation of base bending moment of the frame with perfectly rigid connections at Column C is 31,172 N·m but with the flexible connections of fixity factor $v=0.3$ and damper damping constant $c=4 \times 10^6$ Nsm, it is only 9039 N·m, leading to a 71% reduction.

The overall performance of the example frame under earthquake excitation is also examined in this study. Figs. 10 and 11 show the variations of standard deviation of lateral displacement and shear force, respectively, of the frame with height. It is seen from Fig. 10 that the lateral displacement response of the frame with the flexible connections of $v=0.47$ but without rotational dampers are much larger than the same quantity of the frame of rigid connections at every floor. With the rotational dampers of optimal damping coefficient, the lateral displacement response of the flexible frame of $v=0.47$ is reduced tremendously and becomes much smaller than that of the rigid frame. This reduction occurs not only at the top of the frame but also at all other floor levels. For the shear force response, Fig. 11 demonstrates that the standard deviation of shear force response of the

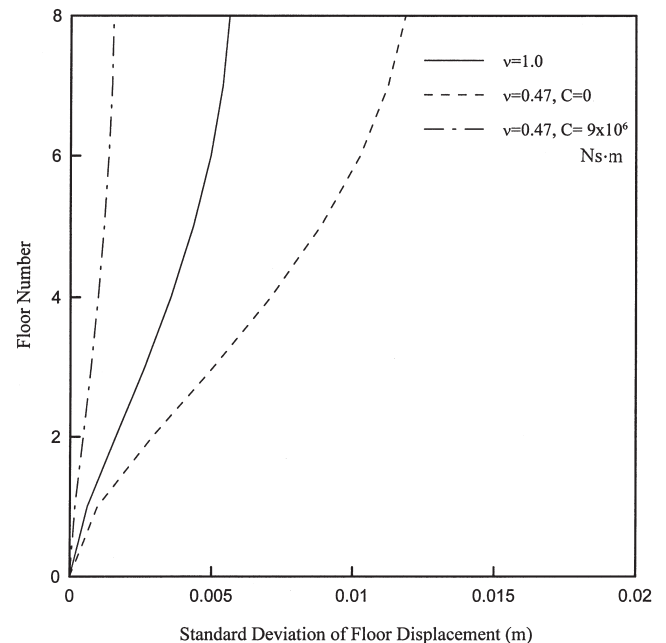


Fig. 10. Variations of lateral displacement response of frame with height.

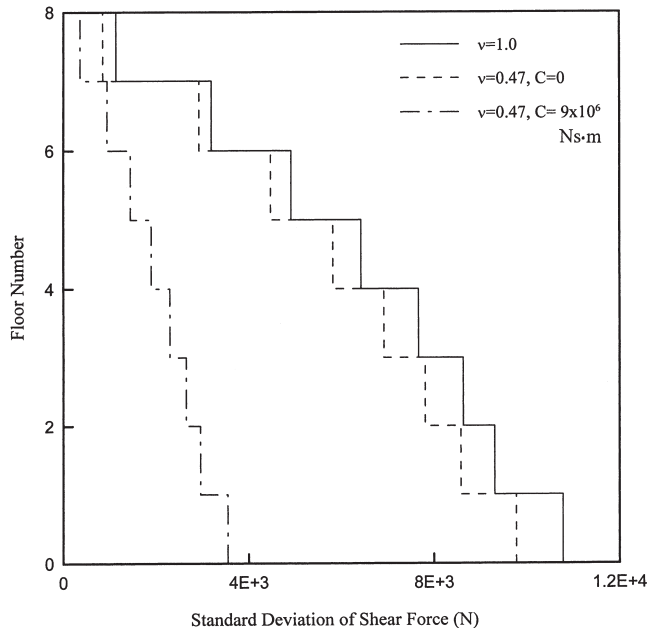


Fig. 11. Variations of shear force response of frame with height.

frame with rigid connections is slightly larger than that of the frame with flexible connections but without rotational dampers. With the rotational dampers of optimal damping coefficient, the shear force response of the flexible frame becomes further smaller at all the levels of the frame. Thus, the overall performance of the example frame is significantly enhanced due to the installation of rotational dampers.

6. Conclusions

The stiffness, mass, and damping matrices of the beam element with rotational spring and damper at its ends have been derived using a combination of the finite element method and direct stiffness method. The solutions for dynamic characteristics and seismic response of the frame with flexible connections and rotational dampers have been also provided in the frequency domain using the complex modal analysis method and the generalized pseudo-excitation method. The formulation was then applied to an example frame and parametric studies were performed. The results showed that there is an optimal damper damping coefficient for a given mode of vibration and a given fixity factor of the frame. With the optimal damper damping coefficient, the modal damping ratio of the frame could be significantly increased and the seismic responses, including lateral displacement, shear force, and bending moment, of the frame could be considerably reduced to the level smaller than those of the frame with rigid connections.

It should be pointed out, however, that while the suggested linear approach in the frequency domain is con-

venient for the determination of modal properties and seismic response of frames with connection dampers, a nonlinear seismic analysis of the frame with nonlinear connection stiffness and geometric effects is worthwhile to be investigated in the time domain in the future. Furthermore, experimental work may be required in the future to verify the achieved results in this study, particularly for the optimal damper damping coefficient and the maximum modal-damping ratio.

Acknowledgements

The writers are grateful for the financial support from the Hong Kong Polytechnic University through its Area of Strategic Development Programme in Structural Vibration Control for the first writer and a PhD studentship to the second writer. They also appreciate the support from Profs J.M. Ko and S.L. Chan, The Department of Civil and Structural Engineering of The Hong Kong Polytechnic University.

Appendix A

The stiffness matrix and the consistent mass matrix of a beam in the finite element method are usually expressed as

$$\mathbf{K}_b = \frac{EI}{L^3} \begin{bmatrix} \frac{AL^2}{I} & 0 & 0 & -\frac{AL^2}{I} & 0 & 0 \\ 0 & 12 & 6L & 0 & -12 & 6L \\ 0 & 6L & 4L^2 & 0 & -6L & 2L^2 \\ -\frac{AL^2}{I} & 0 & 0 & \frac{AL^2}{I} & 0 & 0 \\ 0 & -12 & -6L & 0 & 12 & -6L \\ 0 & 6L & 2L^2 & 0 & -6L & 4L^2 \end{bmatrix} \quad (49)$$

$$\mathbf{M}_b = \frac{\rho AL}{420} \begin{bmatrix} 140 & 0 & 0 & 70 & 0 & 0 \\ 0 & 156 & 22L & 0 & 54 & -13L \\ 0 & 22L & 4L^2 & 0 & 13L & -3L^2 \\ 140 & 0 & 0 & 70 & 0 & 0 \\ 0 & 54 & 13L & 0 & 156 & -22L \\ 0 & -13L & -3L^2 & 0 & -22L & 4L^2 \end{bmatrix} \quad (50)$$

in which E , I , A , ρ , and L are the Young's modulus, second moment of area, cross section area, density and length of beam, respectively.

By re-arranging the order of the elements in the displacement and force vectors in Eqs. (2) and (3), the stiffness matrix of the beam with rotational spring and damper at its ends is

$$\mathbf{K}_e = \frac{2EI}{L} \begin{bmatrix} K_0 & \frac{2(K_{11}+K_{12}+K_{22})}{L^2} & \frac{2K_{11}+K_{12}}{L} & 2K_{11} & 0 & 0 \\ 0 & \frac{2(K_{11}+K_{12}+K_{22})}{L^2} & \frac{2K_{11}+K_{12}}{L} & 0 & \frac{2(K_{11}+K_{12}+K_{22})}{L^2} & 0 \\ 0 & \frac{2K_{11}+K_{12}}{L} & 2K_{11} & 0 & 0 & 0 \\ -K_0 & 0 & 0 & K_0 & 0 & 0 \\ 0 & \frac{2(K_{11}+K_{12}+K_{22})}{L^2} & \frac{2K_{11}+K_{12}}{L} & 0 & \frac{2(K_{11}+K_{12}+K_{22})}{L^2} & 0 \\ 0 & \frac{K_{12}+2K_{22}}{L} & K_{12} & 0 & \frac{K_{12}+2K_{22}}{L} & 2K_{22} \end{bmatrix} \quad \text{Sym.} \quad (51)$$

in which

$$K_0 = \frac{EA}{2EI}, \quad K_{11} = \frac{3v_1}{4-v_1v_2}, \quad K_{12} = \frac{3v_1v_2}{4-v_1v_2}, \quad K_{22} = \frac{3v_2}{4-v_1v_2}$$

v_1 is the fixity factor at node 1 and v_2 is the fixity factor at node 2 defined by Eq. (12).

The damping matrix of the beam with rotational spring and damper at its ends is

$$\mathbf{C}_e = 2 \begin{bmatrix} 0 & \frac{2(C_{11}+C_{12}+C_{22})}{L^2} & \frac{2C_{11}+C_{12}}{L} & 2C_{11} & 0 & 0 \\ 0 & \frac{2(C_{11}+C_{12}+C_{22})}{L^2} & \frac{2C_{11}+C_{12}}{L} & 0 & \frac{2(C_{11}+C_{12}+C_{22})}{L^2} & 0 \\ 0 & \frac{2C_{11}+C_{12}}{L} & 2C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2(C_{11}+C_{12}+C_{22})}{L^2} & \frac{2C_{11}+C_{12}}{L} & 0 & \frac{2(C_{11}+C_{12}+C_{22})}{L^2} & 0 \\ 0 & \frac{C_{12}+2C_{22}}{L} & C_{12} & 0 & \frac{C_{12}+2C_{22}}{L} & 2C_{22} \end{bmatrix} \quad \text{Sym.} \quad (52)$$

where

$$C_{11} = \frac{1}{(4-v_1v_2)^2} [4c_1(1-v_1)^2 + c_2(1-v_2)^2v_1^2]$$

$$C_{12} = \frac{4}{(4-v_1v_2)^2} [c_1v_2(1-v_1)^2 + c_2v_1(1-v_2)^2]$$

$$C_{22} = \frac{1}{(4-v_1v_2)^2} [4c_2(1-v_2)^2 + c_1(1-v_1)^2v_2^2]$$

The mass matrix of the beam with rotational damper and spring at its ends is the summation of two parts $\mathbf{M}_k$ and $\mathbf{M}_m$. $\mathbf{M}_k$ includes only the fixity factors, and for the case of rigid connections it coincides with the consistent mass matrix of the beam expressed by Eq. (50). $\mathbf{M}_m$ includes both the fixity factors and the damper damping coefficients.

$$\mathbf{M}_e = \mathbf{M}_k + \mathbf{M}_m =$$

$$\frac{\rho AL}{420} \begin{bmatrix} 140 & & & & & \\ 0 & 4Z_1(v_1, v_2) & & & & \\ 0 & 2LZ_2(v_1, v_2) & 4L^2Z_3(v_1, v_2) & & & \\ 70 & 0 & 0 & 140 & & \\ 0 & 2Z_3(v_1, v_2) & LZ_4(v_2, v_1) & 0 & 4Z_1(v_2, v_1) & \\ 0 & -LZ_4(v_1, v_2) & -L^2Z_6(v_1, v_2) & 0 & -2LZ_3(v_2, v_1) & 4L^2Z_5(v_2, v_1) \end{bmatrix} \quad \text{Sym.} \quad (53)$$

$$+ \frac{2L}{EI} \begin{bmatrix} 0 & \frac{2(M_{11}+M_{12}+M_{22})}{L^2} & \frac{2M_{11}+M_{12}}{L} & 2M_{11} & 0 & 0 \\ 0 & \frac{2(M_{11}+M_{12}+M_{22})}{L^2} & \frac{2M_{11}+M_{12}}{L} & 0 & \frac{2(M_{11}+M_{12}+M_{22})}{L^2} & 0 \\ 0 & \frac{2M_{11}+M_{12}}{L} & 2M_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2(M_{11}+M_{12}+M_{22})}{L^2} & \frac{2M_{11}+M_{12}}{L} & 0 & \frac{2(M_{11}+M_{12}+M_{22})}{L^2} & 0 \\ 0 & \frac{M_{12}+2M_{22}}{L} & M_{12} & 0 & \frac{M_{12}+2M_{22}}{L} & 2M_{22} \end{bmatrix} \quad \text{Sym.}$$

where

$$Z_1(p, q) = \frac{1}{(4-pq)^2} (560 + 224p - 196q - 328pq + 32p^2 + 32q^2 + 50pq^2 - 55p^2q + 32p^2q^2)$$

$$Z_2(p, q) = \frac{p}{(4-pq)^2} (224 + 64p - 160q - 86pq + 32q^2 + 25pq^2)$$

$$Z_3(p, q) = \frac{1}{(4-pq)^2} (560 - 28p - 28q - 184pq - 64p^2 - 64q^2 + 5pq^2 + 5p^2q + 41p^2q^2)$$

$$Z_4(p, q) = \frac{q}{(4-pq)^2} (392 - 100p - 128q - 38pq - 64p^2 + 55p^2q)$$

$$Z_5(p, q) = \frac{p^2}{(4-pq)^2} (32 - 31q + 8q^2)$$

$$Z_6(p, q) = \frac{pq}{(4-pq)^2} (124 - 64p - 64q + 31pq)$$

$$M_{11} = \frac{1}{3(4-v_1v_2)^3} [8c_1c_2v_1(1-v_1)^2(1-v_2)^2 - 4c_1^2(1-v_1)^3(4-v_2) - c_2^2v_1^2(1-v_2)^3(4-v_1)]$$

$$M_{12} = \frac{4}{3(4-v_1v_2)^3} [c_1c_2(4+v_1v_2)(1-v_1)^2(1-v_2)^2 - c_1^2(1-v_1)^3(4-v_2)v_2 - c_2^2(1-v_2)^3(4-v_1)v_1]$$

$$M_{22} = \frac{1}{3(4-v_1v_2)^3} [8c_1c_2v_2(1-v_1)^2(1-v_2)^2 - 4c_2^2(1-v_2)^3(4-v_1) - c_1^2v_2^2(1-v_1)^3(4-v_2)]$$

It is noted that if the axial effects are neglected, the matrices $\mathbf{K}_e$, $\mathbf{C}_e$ and $\mathbf{M}_e$ given in Eqs. (51) to (53) and derived in terms of the combination of the finite element method and the direct stiffness method are the same as those given by Kawashima and Fujimoto [8] using the direct stiffness method only.

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